

Oracle E-Business Suite GitOps Modernization

MVP1 Architecture and Delivery Status

Field	Value
Version	1.4
Date	2026-10-06
Status	Proposed
EBS	12.2.12
Database	Oracle Database 19c Enterprise Edition Extreme Performance
Environment reviewed	DEV

1. Purpose

Define a phased process to inventory, detect, capture, review, and eventually deploy custom Oracle E-Business Suite changes using GitHub.

The target solution will not use AIFO, Jenkins, or SVN.

MVP1 has three sequential parts:

- 1. MVP1-A — Inventory and detection
- 2. MVP1-B — Capture and manifest
- 3. MVP1-C — Controlled deployment to DEV

Automated deployment is excluded from MVP1-A and MVP1-B.

2. Environment Findings

The following findings come from the database evidence provided for DEV:

Area	Finding
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EBS release	12.2.12
Database	Oracle Database 19c EE Extreme Performance
Container	DEV
Current session edition	V_20261001_1432
Edition status	V_20261001_1432 is listed as usable in DBA_EDITIONS
Recent ADOP record	Session 426 records a cutover phase on 01-Oct-2026
Custom objects	XX% objects appear under APPS and multiple custom schemas
Developer accounts	Individual DEV_* accounts are present
Direct APPS sessions	SQL Developer sessions as APPS appear in the provided session results
Broad privileges	DEVELOPER_ROLE and DNVGL_DEVELOPER_ROLE have broad CREATE ANY and ALTER ANY privileges
Unified Auditing	The Unified Auditing option is reported as enabled
Existing DDL trigger	XXISV.XXISV_MSFTTR_CTL_DDLH is enabled for create, alter, or drop events
Audit evidence	The supplied audit samples do not establish complete DDL coverage for all custom schemas

The current edition being usable does **not**, by itself, confirm the run edition. The ADOP row is historical evidence of a cutover phase, not proof of the current ADOP status. Both must be verified using an approved operational procedure before capture or deployment.

3. Prerequisites and Governance

These are readiness requirements, not completed project deliverables. Track each as **Not assessed**, **In progress**, or **Completed**, and retain evidence.

3.1 Developer accounts and schema ownership

- Inventory developer accounts, shared accounts, proxy connections, and connections made as APPS.
- Require developers to use individually attributable accounts.

- Require routine custom development to take place in each developer's assigned schema, not in APPS or another shared schema.
- Document and approve exceptions where an object must be owned by APPS or another shared schema.
- Define how cross-schema access is granted without providing unnecessary privileges.

3.2 Privilege review

- Review direct grants, role grants, and effective privileges for developer accounts and roles.
- Assess broad privileges, including CREATE ANY and ALTER ANY.
- Determine the minimum privileges required for approved development activities.
- Restrict excessive privileges through a controlled DBA and security process.
- Document approved exceptions and retain evidence of the review.

Privilege changes must not be performed automatically by the capture or deployment tool.

3.3 Audit and change attribution

- Assess Unified Audit policies and existing triggers for coverage of all approved schemas.
- Define how DDL changes will be attributed to individual users, including proxy connections and approved shared-account exceptions.
- Define audit retention, access, and review requirements.
- Treat hashes as evidence that an object changed, not as proof of who changed it.

3.4 Operational readiness

- Confirm the run edition and the supported ADOP status check.
- Define conditions that block inventory, capture, or deployment.
- Approve the schema and object-type scope.
- Define separate least-privilege identities for capture and deployment.
- Approve storage for snapshots, logs, and release artifacts.

- Establish GitHub access, review, and secret-management controls.

3.5 Readiness tracker

Prerequisite	Status	Owner
Developer account and connection inventory	Not assessed	DBA / Security
Policy requiring assigned schemas for routine development	Not assessed	EBS owner / DBA
Shared-schema exceptions documented	Not assessed	EBS owner / DBA
Effective developer privileges reviewed	Not assessed	DBA / Security
Excessive privileges assessed and remediation planned	Not assessed	DBA / Security
DDL audit coverage and retention assessed	Not assessed	DBA / Security
Run edition and ADOP check verified	Not assessed	EBS DBA
Capture and deployment identities defined	Not assessed	DBA / Security
Initial scope approved	Not assessed	EBS owner / DBA
GitHub and secret-management controls approved	Not assessed	Platform / Security

4. MVP1-A — Inventory and Detection

Goal

Create a repeatable inventory of custom objects and identify changed candidates without deploying changes.

Deliverables

- Inventory of objects in the approved scope.
- Classification by owner and object type.

- Identification of editioned and noneditioned objects.
- Capture of the verified run edition.
- Normalized hashes for PL/SQL source.
- Candidate report: NEW, MODIFIED, MISSING, and UNCHANGED.
- Dependency report, including unresolved dependencies.
- Design for identifying the database session responsible for future DDL changes.

Acceptance criteria

- Repeated snapshots of unchanged objects produce stable hashes.
- A controlled change is detected and classified correctly.
- Run edition and ADOP pre-check procedures are verified and documented.
- Edition classification is tested against the DEV database.
- Dependency results include resolution status.
- The session-attribution design is reviewed by the DBA and security team.
- No deployment or DDL execution is performed.

5. MVP1-B — Capture and Manifest

Goal

Let a user select candidates, extract their source, create a manifest, and submit the change for review in GitHub.

Deliverables

- User selection of candidate objects through an agreed change-set mechanism.
- Source extraction for approved object types.
- Manifest with object identity, owner, dependencies, selection origin, and SHA-256 checksums.

- Automated manifest and checksum validation.
- GitHub branch creation and pull request submission.
- Artifact publication only after approval.

Acceptance criteria

- Selected objects are extracted into the agreed repository structure.
- The manifest passes validation.
- A pull request is created with source and manifest.
- Approval is recorded before artifact publication.
- The process is repeatable and traceable.
- No deployment or DDL execution is performed.

6. MVP1-C — Controlled DEV Deployment

Goal

Deploy an approved, immutable release to DEV through a private runner, with ADOP controls, validation, and rollback.

Deliverables

- Private runner within the DEV environment.
- Controlled ADOP-aware execution.
- Preflight checks.
- Application of the release in DEV.
- Compilation and post-deployment validation.
- Deployment history.
- Rollback by redeploying a previous release.

Acceptance criteria

- Deployment passes documented preflight checks.
- Compilation and post-deployment validations are recorded.
- Each deployment and execution step is traceable.
- Failures stop execution and are recorded.
- Rollback to a previous release is tested.
- The exact ADOP procedure is validated against the target EBS environment before implementation.

7. Scope

7.1 Initial object types

Subject to approval and tested extraction rules:

- Packages and package bodies
- Procedures and functions
- Views
- Triggers
- Types and type bodies
- Synonyms
- Sequences
- Tables and indexes
- Grants, if explicitly included

AOL objects, Forms, Reports, OAF, BI Publisher, and Workflow artifacts are outside the initial database-object scope. They may be considered in a later phase with appropriate extraction methods.

7.2 Initial schema allowlist

Retain the current draft allowlist for now. The custom schema entries remain placeholders until their names are approved.

```
scope:
  schemas:
    - APPS
    - <APPROVED_CUSTOM_SCHEMA_1>
    - <APPROVED_CUSTOM_SCHEMA_2>

  object_name_prefixes:
    - XX

  excluded_schemas:
    - SYS
    - SYSTEM

  excluded_object_name_patterns:
    - APEX_%
    - WWV_%
```

The allowlist must be explicit. It must not be replaced by a scan of every schema.

8. Detection and Dependencies

- Detect changes using normalized content hashes, not `LAST_DDL_TIME` alone.
- Use `LAST_DDL_TIME` only as an optional way to narrow the objects examined.
- Report missing objects as candidates for review; do not generate deletion scripts automatically.
- Use `DBA_DEPENDENCIES` as a technical source, not as a complete EBS dependency model.
- Identify dependencies outside the approved scope and flag unresolved dependencies.
- Do not automatically include Oracle-delivered objects.

Dependency status values:

- `RESOLVED_IN_CHANGESET`
- `RESOLVED_IN_REPOSITORY`
- `EXISTS_IN_SOURCE_ENVIRONMENT`
- `UNRESOLVED`

- `EXTERNAL_ORACLE_OBJECT`

9. Manifest and Release Controls

The MVP1-B manifest describes the release; MVP1-C may add execution metadata.

Minimum manifest information:

- Release and change identifiers
- Source environment and captured edition
- Object owner, name, type, and edition status
- Source path and SHA-256 checksum
- Whether an object was explicitly selected or included as a dependency
- Dependencies and resolution status
- Validation result and warnings

An unresolved dependency prevents the manifest from being marked fully validated. Artifacts are immutable after approval.

10. Security and Access Model

Capture: MVP1-A and MVP1-B

- Use a dedicated, read-only database identity where feasible.
- Grant only access needed for the approved scope.
- Use narrowly scoped GitHub credentials.
- Store secrets in an approved secret manager.
- Do not place credentials in logs, manifests, command lines, or repositories.

Deployment: MVP1-C

- Use a separate least-privilege deployment identity.

- Do not use developer accounts or a shared APPS password as the general deployment identity.
- Separate credentials by environment.
- Require explicit approval for operations that modify EBS objects.
- Do not give GitHub-hosted runners direct database access.

11. Current Status

Confirmed from supplied evidence

- EBS release is 12.2.12.
- Database is Oracle Database 19c EE Extreme Performance.
- The database container is DEV.
- Current session edition is V_20261001_1432; it is listed as usable.
- The inventory query shows XX% objects across APPS and multiple custom schemas.
- The supplied privilege query lists broad privileges for DEVELOPER_ROLE and DNVGL_DEVELOPER_ROLE.
- The supplied session query includes SQL Developer sessions as APPS.
- Unified Auditing is reported as enabled.
- An enabled DDL trigger exists in XXISV.

Still requiring validation or implementation

- Verify that V_20261001_1432 is the run edition.
- Verify the current ADOP status and approved blocking conditions.
- Review developer accounts, role membership, and effective privileges.
- Establish and enforce the assigned-schema development policy.
- Assess audit coverage across all approved custom schemas.
- Approve the schema and object-type allowlists.

- Define capture and deployment identities.
- Implement and test the MVP1-A, MVP1-B, and MVP1-C deliverables.

12. Roadmap

Phase	Outcome
Governance prerequisites	Developer schema policy, privilege review, audit assessment, approved scope
MVP1-A	Inventory, edition classification, PL/SQL hashes, candidates, session-attribution design, dependencies
MVP1-B	User-selected capture, source extraction, manifest, GitHub pull request, approved artifact
MVP1-C	Private runner, controlled ADOP, DEV deployment, validation, history, rollback
Later phases	Additional environments and non-database EBS artifacts

13. Open Decisions

- Final custom schema names for the allowlist.
- Supported run-edition verification procedure.
- Supported ADOP status check and blocking conditions.
- Snapshot and log storage.
- Capture and deployment identities and privileges.
- Change-selection mechanism.
- DDL audit and session-attribution design.
- GitHub repository and approval configuration.
- Approved secret manager.
- Owner schema for deployment-history tables.